

Learned closures and nonlinear manifolds for parametric projection-based reduced models

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September 10, 2026

Abstract

We compare learned closures and nonlinear manifolds for parametric projection-based reduced-order models of a nonlinear time-dependent system. All models use a common Euclidean proper-orthogonal-decomposition (POD) coordinate system and are trained from full-residual linear projection-based reduced-order model (PROM) trajectories. The comparison includes state-conditioned, parameter-time-conditioned, and hybrid PROM-ANN closures; an intrusive PROM-POD-AE decoder manifold; a non-intrusive POD-NN-ROM direct coefficient map; and a non-intrusive latent POD-DL-ROM.

The numerical study uses a parametric two-dimensional inviscid Burgers benchmark. The intrusive models all minimize the same time-discrete residual, but over different learned trial maps. A shared parameter-time map gives a controlled sweep from a non-intrusive POD-NN-ROM endpoint to the full-coordinate linear PROM. A complementary tail-perturbation diagnostic quantifies how a prescribed Case-2 secondary-coordinate error transfers to the residual-solved primary coordinates. A nested coefficient-data study with 9, 9 + 8, and 9 + 18 PROM trajectories separates in-domain improvement from extrapolation behaviour. We also state the empirical cubature method (ECM) formulation required for a future HPROM extension; all numerical results reported here are full-residual PROM results.

1 Introduction

Projection-based model order reduction (PMOR) reduces a high-dimensional model while retaining its governing residual and its parameter dependence [1, 2]. For nonlinear, transport-dominated systems, an accurate linear subspace may nevertheless require many coordinates. Nonlinear trial manifolds, learned closures, and latent representations seek to reduce the number of online unknowns without discarding the projection principle [3, 4, 5].

This work compares intrusive closure-enhanced PROMs (PROM-ANN), an intrusive decoder manifold (PROM-POD-AE), a non-intrusive direct coefficient map (POD-NN-ROM), and a non-intrusive latent map (POD-DL-ROM). PROM means that the full time-discrete residual is minimized online. It is not synonymous with a generic ROM: the non-intrusive baselines do not evaluate a residual online. Likewise, the HPROM formulation below is a sampled-residual extension of PROM, not a second name for the same computation.

The reported campaign uses a standard Euclidean POD basis and a common full-residual linear PROM teacher. This permits a direct comparison between Case 2, which prescribes a learned secondary-coordinate tail but solves a primary coordinate block, and POD-NN-ROM, which predicts the full coordinate vector from the same parameter-time map. Hyperreduction is developed explicitly because an ECM rule must be tied to the trial manifold and its LSPG tangent; HPROM numerical results are deliberately deferred until the corresponding campaign is available.

2 Projection Framework

2.1 Time-discrete residual and Euclidean POD basis

Consider a parametric nonlinear semi-discrete initial-value problem

$$\mathbf{M}(\boldsymbol{\mu})\dot{\mathbf{u}}(t; \boldsymbol{\mu}) + \mathbf{f}(\mathbf{u}(t; \boldsymbol{\mu}); \boldsymbol{\mu}) - \mathbf{g}(t; \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{u}(0; \boldsymbol{\mu}) = \mathbf{u}_0(\boldsymbol{\mu}), \quad (1)$$

with $\boldsymbol{\mu} \in \mathcal{D}$. After a chosen implicit time discretization, let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the residual at step $m+1$, with dependence on the previous state understood. The HDM solves

$$\mathbf{r}^{m+1}(\mathbf{u}^{m+1}; \boldsymbol{\mu}) = \mathbf{0}. \quad (2)$$

Let $\mathbf{u}_{\text{ref}}$ be the snapshot mean and let $\mathbf{S}_c$ collect the centered HDM snapshots. The thin singular-value decomposition

$$\mathbf{S}_c = \mathbf{U}\boldsymbol{\Sigma}\mathbf{Y}^T \quad (3)$$

defines the Euclidean POD basis retained in this work,

$$\mathbf{V}_{\text{tot}} = \mathbf{U}(:, 1 : n_{\text{tot}}), \quad \mathbf{V}_{\text{tot}}^T \mathbf{V}_{\text{tot}} = \mathbf{I}. \quad (4)$$

The common affine linear trial manifold is

$$\phi_N(\mathbf{q}_N) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}}\mathbf{q}_N, \quad \mathbf{q}_N \in \mathbb{R}^{n_{\text{tot}}}. \quad (5)$$

All learned trial maps below are expressed in these same coordinates.

2.2 Linear PROM and LSPG reference

The full-residual linear PROM is the time-discrete least-squares Petrov–Galerkin (LSPG) problem

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\mathbf{r}^{m+1}(\phi_N(\mathbf{x}); \boldsymbol{\mu})\|_2^2. \quad (6)$$

At Gauss–Newton iterate k , define

$$\mathbf{T}_N = \mathbf{V}_{\text{tot}}, \quad \mathbf{J}^{m+1,k} = \left. \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}} \right|_{\phi_N(\mathbf{q}_N^{m+1,k})}, \quad \mathbf{W}_N^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_N. \quad (7)$$

The normal equations for the increment are

$$\left(\mathbf{W}_N^{m+1,k}\right)^T \mathbf{W}_N^{m+1,k} \Delta \mathbf{q}_N^{m+1,k} = -\left(\mathbf{W}_N^{m+1,k}\right)^T \mathbf{r}^{m+1}\left(\phi_N(\mathbf{q}_N^{m+1,k}); \boldsymbol{\mu}\right). \quad (8)$$

Equation (6) generates all training targets in the PROM campaign.

2.3 Linear HPROM and ECM formulation

The linear HPROM replaces full residual evaluations in the LSPG update by a sparse, positive empirical-cubature approximation. This formulation is given in detail because the same projected-residual algebra is used when the affine trial manifold is replaced by a learned one. Let $\mathcal{E} = \{e_i\}_{i=1}^{N_e}$ be the mesh entities. The time-discrete residual is assembled from local contributions as

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ \mathbf{u}; \boldsymbol{\mu}). \quad (9)$$

Here $\mathbf{r}_{e_i}^{m+1} \in \mathbb{R}^{d_{e_i}}$ is the contribution of entity e_i , $\mathbf{L}_{e_i} \in \{0, 1\}^{d_{e_i} \times N}$ extracts its residual degrees of freedom, and $\mathbf{L}_{e_i}^+$ extracts the enlarged state stencil needed to evaluate it. The stencil contains neighbouring entities whenever the discretization requires them.

For the affine trial map (5), the projected LSPG residual at a Gauss–Newton iterate has the entity-wise form

$$\begin{aligned}\mathbf{g}_N^{m+1,k}(\boldsymbol{\mu}) &:= \left(\mathbf{W}_N^{m+1,k}\right)^T \mathbf{r}^{m+1}\left(\phi_N(\mathbf{q}_N^{m+1,k}); \boldsymbol{\mu}\right) \\ &= \sum_{e_i \in \mathcal{E}} \left(\mathbf{L}_{e_i} \mathbf{W}_N^{m+1,k}\right)^T \mathbf{r}_{e_i}^{m+1}\left(\mathbf{L}_{e_i}^+ \phi_N(\mathbf{q}_N^{m+1,k}); \boldsymbol{\mu}\right).\end{aligned}\quad (10)$$

Let $\mathbf{J}_{e_i}^{m+1,k}$ be the derivative of the local residual with respect to the stencil state at the same iterate. The Gauss–Newton matrix is assembled analogously,

$$\mathbf{A}_N^{m+1,k} := \left(\mathbf{W}_N^{m+1,k}\right)^T \mathbf{J}^{m+1,k} \mathbf{T}_N = \sum_{e_i \in \mathcal{E}} \left(\mathbf{L}_{e_i} \mathbf{W}_N^{m+1,k}\right)^T \mathbf{J}_{e_i}^{m+1,k} \left(\mathbf{L}_{e_i}^+ \mathbf{T}_N\right). \quad (11)$$

ECM selects a reduced entity set $\tilde{\mathcal{E}}_N \subset \mathcal{E}$ and positive weights ξ_{N,e_i} . The same rule approximates both projected quantities,

$$\begin{aligned}\hat{\mathbf{g}}_N^{m+1,k} &= \sum_{e_i \in \tilde{\mathcal{E}}_N} \xi_{N,e_i} \left(\mathbf{L}_{e_i} \mathbf{W}_N^{m+1,k}\right)^T \mathbf{r}_{e_i}^{m+1}\left(\mathbf{L}_{e_i}^+ \phi_N(\mathbf{q}_N^{m+1,k}); \boldsymbol{\mu}\right), \\ \hat{\mathbf{A}}_N^{m+1,k} &= \sum_{e_i \in \tilde{\mathcal{E}}_N} \xi_{N,e_i} \left(\mathbf{L}_{e_i} \mathbf{W}_N^{m+1,k}\right)^T \mathbf{J}_{e_i}^{m+1,k} \left(\mathbf{L}_{e_i}^+ \mathbf{T}_N\right).\end{aligned}\quad (12)$$

The online sampled update is consequently

$$\hat{\mathbf{A}}_N^{m+1,k} \Delta \mathbf{q}_N^{m+1,k} = -\hat{\mathbf{g}}_N^{m+1,k}, \quad \mathbf{q}_N^{m+1,k+1} = \mathbf{q}_N^{m+1,k} + \Delta \mathbf{q}_N^{m+1,k}. \quad (13)$$

Offline cubature construction. For every selected parameter–time training state s , the linear coordinate $\mathbf{q}_{N,s}$ and its LSPG test basis $\mathbf{W}_{N,s}$ define the contribution of entity e_i to the projected residual,

$$\mathbf{c}_{N,e_i}^{(s)} = (\mathbf{L}_{e_i} \mathbf{W}_{N,s})^T \mathbf{r}_{e_i}(\mathbf{L}_{e_i}^+ \phi_N(\mathbf{q}_{N,s}); \boldsymbol{\mu}_s). \quad (14)$$

Stacking all n_{tot} -component contributions over N_{ECM} selected states gives

$$\mathbf{C}_N = \begin{bmatrix} \mathbf{c}_{N,e_1}^{(1)} & \cdots & \mathbf{c}_{N,e_{N_e}}^{(1)} \\ \vdots & & \vdots \\ \mathbf{c}_{N,e_1}^{(N_{\text{ECM}})} & \cdots & \mathbf{c}_{N,e_{N_e}}^{(N_{\text{ECM}})} \end{bmatrix} \in \mathbb{R}^{n_{\text{tot}} N_{\text{ECM}} \times N_e}, \quad \mathbf{b}_N = \mathbf{C}_N \mathbf{1}. \quad (15)$$

ECM seeks a sparse nonnegative approximation of the full projected-residual data,

$$\boldsymbol{\xi}_N = \arg \min_{\boldsymbol{\xi} \geq \mathbf{0}} \|\mathbf{C}_N \boldsymbol{\xi} - \mathbf{b}_N\|_2^2, \quad \tilde{\mathcal{E}}_N = \{e_i \in \mathcal{E} : \xi_{N,e_i} > 0\}. \quad (16)$$

In the present implementation, a direct SVD removes numerically null directions of $\mathbf{C}_N$ before the nonnegative cubature solve. The rule is therefore tied to the affine trial manifold and its LSPG test basis. In particular, replacing ϕ_N by a learned closure or decoder also changes the tangent, the test basis, and the projected contributions; the corresponding ECM rule must be constructed for that map.

Relation to other hyperreduction methods. The common step is to approximate an entity-wise sum of *projected* residual contributions, rather than the state reconstruction or a learned map. We use ECM in the sense of empirical cubature [6, 7]. ECSW is another project–then–approximate hyperreduction strategy [8]. It could be used with the same LSPG and trial manifold formulations, but ECM is the rule used in this work.

3 Learned Reduced Representations

3.1 PROM–ANN closures

Partition the common POD basis and coordinate vector as

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q}_p \\ \mathbf{q}_s \end{bmatrix}, \quad n + \bar{n} = n_{\text{tot}}. \quad (17)$$

PROM–ANN supplies the secondary block through a learned relation

$$\mathbf{q}_s = \mathcal{F}(\mathbf{q}_p, \boldsymbol{\mu}, t), \quad \phi_{\mathcal{F}}(\mathbf{q}_p; \boldsymbol{\mu}, t) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}_p + \bar{\mathbf{V}}\mathcal{F}(\mathbf{q}_p, \boldsymbol{\mu}, t). \quad (18)$$

The network provides reduced secondary coordinates, not a direct HDM state approximation. The primary coordinates remain online unknowns of a residual minimization. This is the neural-network-augmented projection construction of [3]; analogous closure roles have also been represented by non-neural regressors [9].

The three cases are information patterns in this common construction:

$$\text{Case 1: } \mathbf{q}_s = \mathcal{N}(\mathbf{q}_p), \quad \text{Case 2: } \mathbf{q}_s = \mathcal{M}(\boldsymbol{\mu}, t), \quad \text{Case 3: } \mathbf{q}_s = \mathcal{H}(\mathbf{q}_p, \boldsymbol{\mu}, t). \quad (19)$$

Case 1 is state-conditioned, Case 2 parameter–time-conditioned, and Case 3 combines both inputs. The LSPG problem is

$$\mathbf{q}_p^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\phi_{\mathcal{F}}(\mathbf{x}; \boldsymbol{\mu}, t^{m+1}); \boldsymbol{\mu})\|_2^2, \quad (20)$$

with tangent

$$\mathbf{T}_{\mathcal{F}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}_p}, \quad \mathbf{W}_{\mathcal{F}} = \mathbf{J} \mathbf{T}_{\mathcal{F}}. \quad (21)$$

For Case 2, $\partial \mathcal{M} / \partial \mathbf{q}_p = \mathbf{0}$ and $\mathbf{T}_{\mathcal{M}} = \mathbf{V}$. Its tangent is therefore the cheapest, but its prescribed secondary block is not corrected through a state-dependent tangent.

3.2 PROM–POD–AE and intrusive trial maps

An autoencoder maps the complete linear coordinate vector through a latent variable,

$$\mathbf{z} = \mathcal{E}(\mathbf{q}_N), \quad \hat{\mathbf{q}}_N = \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{n_z}. \quad (22)$$

PROM–POD–AE uses the decoder as an intrusive nonlinear trial manifold,

$$\phi_{\text{AE}}(\mathbf{z}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z}), \quad \mathbf{T}_{\text{AE}} = \mathbf{V}_{\text{tot}} \frac{\partial \mathcal{D}}{\partial \mathbf{z}}. \quad (23)$$

Its latent coordinate minimizes the same residual as (20), but over $\mathbf{z}$ rather than $\mathbf{q}_p$. This is distinct from POD–DL–ROM, which is non-intrusive.

Table 1: Trial decoders and tangent bases of the intrusive models. Every row minimizes the same time-discrete residual over its displayed online variable.

Model	Online variable	Trial decoder	Tangent
Linear PROM/HPROM	$\mathbf{q}_N$	$\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N$	$\mathbf{V}_{\text{tot}}$
PROM–ANN Case 1	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}_p + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}_p)$	$\mathbf{V} + \bar{\mathbf{V}} \partial \mathcal{N} / \partial \mathbf{q}_p$
PROM–ANN Case 2	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}_p + \bar{\mathbf{V}}\mathcal{M}(\boldsymbol{\mu}, t)$	$\mathbf{V}$
PROM–ANN Case 3	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}_p + \bar{\mathbf{V}}\mathcal{H}(\mathbf{q}_p, \boldsymbol{\mu}, t)$	$\mathbf{V} + \bar{\mathbf{V}} \partial \mathcal{H} / \partial \mathbf{q}_p$
PROM–POD–AE	$\mathbf{z}$	$\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z})$	$\mathbf{V}_{\text{tot}} \partial \mathcal{D} / \partial \mathbf{z}$

3.3 Model-specific ECM construction

The HPROM extension does not reuse the linear rule in (16). For any intrusive map ϕ_c and tangent $\mathbf{T}_c$, the entity contributions use $\mathbf{W}_c = \mathbf{J}\mathbf{T}_c$. A manifold coordinate is first fitted at every selected HDM state,

$$\mathbf{y}_{c,s} = \arg \min_{\mathbf{y}} \left\| \mathbf{u}_s^{m_s+1} - \phi_c(\mathbf{y}; \boldsymbol{\mu}_s, t^{m_s+1}) \right\|_2^2. \quad (24)$$

For Case 2 this is linear least squares after subtracting the prescribed tail,

$$\mathbf{q}_{p2,s} = \mathbf{V}^\dagger \left[\mathbf{u}_s^{m_s+1} - \mathbf{u}_{\text{ref}} - \overline{\mathbf{V}}\mathcal{M}(\boldsymbol{\mu}_s, t^{m_s+1}) \right]. \quad (25)$$

For Cases 1 and 3 and for the decoder manifold, it is nonlinear least squares, for example with the Gauss–Newton update

$$\mathbf{y}_{c,s}^{(k+1)} = \mathbf{y}_{c,s}^{(k)} - \left[\frac{\partial \phi_c}{\partial \mathbf{y}} \right]^\dagger \left[\phi_c(\mathbf{y}_{c,s}^{(k)}; \boldsymbol{\mu}_s, t^{m_s+1}) - \mathbf{u}_s^{m_s+1} \right]. \quad (26)$$

At the fitted state,

$$\mathbf{c}_{c,e_i}^{(s)} = (\mathbf{L}_{e_i} \mathbf{W}_{c,s})^T \mathbf{r}_{e_i}(\mathbf{L}_{e_i}^\dagger \phi_c(\mathbf{y}_{c,s}; \boldsymbol{\mu}_s, t^{m_s+1}); \boldsymbol{\mu}_s). \quad (27)$$

Stacking these vectors produces $\mathbf{C}_c$ and $\mathbf{b}_c = \mathbf{C}_c \mathbf{1}$ as in (16); a positive ECM rule is then computed for each map. The trained closure or decoder, its tangent, and the parameter–time sample set are therefore part of a rule. A linear-HPROM rule, or one built for a different nonlinear map, cannot be reused [6, 7, 10].

3.4 Non-intrusive reference models

POD–NN–ROM is a direct map

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{G}_q(\boldsymbol{\mu}, t), \quad (28)$$

and does not solve a residual online. POD–DL–ROM uses a learned latent map and decoder,

$$\mathbf{z}(t; \boldsymbol{\mu}) \approx \mathcal{G}_z(\boldsymbol{\mu}, t), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)), \quad (29)$$

following the POD–DL–ROM principle [11]. These are direct reduced-coordinate regressors, not PROM residual solves.

4 Benchmark and PROM Protocol

4.1 Burgers problem and discretization

We use the two-dimensional parametric inviscid Burgers benchmark of [3],

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (30)$$

where

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x^2 \\ \frac{1}{2} u_x u_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x u_y \\ \frac{1}{2} u_y^2 \end{bmatrix}, \quad \mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}. \quad (31)$$

The domain is $[0, 100]^2$, $T_f = 25$, and $\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]$. The HDM uses a cell-centered finite-volume Godunov discretization on a 250×250 grid ($N = 125,000$) and 500 uniform time steps with $\Delta t = 0.05$.

4.2 Data, validation, and online points

Nine HDM trajectories on a 3×3 tensor grid define the centered snapshot matrix for the Euclidean POD. We retain $n_{\text{tot}} = 151$ modes, which capture 99.9900% of its energy. At these same nine parameters we solve the full-residual linear PROM. The resulting PROM-generated trajectories, rather than HDM snapshots orthogonally projected after the fact, are the learning targets for every model. This comparison-control choice aligns every learned model with the reduced optimality problem used as its teacher; it is not the conventional purely non-intrusive workflow, which typically uses projected HDM labels.

Two complete 151-coordinate linear-PROM trajectories are reserved before training,

$$\boldsymbol{\mu}^{\text{val},1} = (4.5625, 0.02625), \quad \boldsymbol{\mu}^{\text{val},2} = (5.1875, 0.01875). \quad (32)$$

Because each held-out parameter contains all time samples, this is a parameter-generalization test rather than a random time-snapshot split. The four online evaluations are

$$\begin{aligned} \boldsymbol{\mu}^{(v)} &= (4.875, 0.0225), & \boldsymbol{\mu}^{(1)} &= (4.560, 0.0190), \\ \boldsymbol{\mu}^{(2)} &= (5.190, 0.0260), & \boldsymbol{\mu}^{(3)} &= (4.000, 0.0330). \end{aligned} \quad (33)$$

The first three are in-domain and $\boldsymbol{\mu}^{(3)}$ is extrapolatory. The validation locations are neither training nor online evaluation points.

4.3 Nested coefficient-data enrichment

To isolate data coverage from architecture and online formulation, we use a fixed seed-42 Latin-hypercube candidate design over the original parameter box and a 25% coordinate-wise margin. Two deterministic nested subsets are retained: $9 + 8$ contains the baseline grid, four interior points, and four margin points; $9 + 18$ contains the same baseline and early points plus five additional interior and five additional margin points. The complete $9 + 36$ candidate design defines the nested ordering only; no $9 + 36$ PROM coefficient campaign is used in this study. All online evaluation and parameter-validation locations are excluded from the enrichment sets.

The word “margin” denotes the union of the left, right, top, and bottom strips of the expanded rectangle. It does not enforce coverage of their intersections. This distinction matters for the upper-left extrapolation point $\boldsymbol{\mu}^{(3)}$ and will be revisited with the online results.

Table 2: Nested Euclidean-POD PROM coefficient data. The same two complete linear-PROM trajectories are held out for parameter validation at every level.

Quantity	Baseline	Early	Intermediate
Baseline grid trajectories	9	9	9
Interior LHS trajectories	0	4	9
Margin LHS trajectories	0	4	9
Total training trajectories	9	17	27
Training samples	4509	8517	13527
Held-out parameter-validation trajectories	2	2	2
Held-out parameter-validation samples	1002	1002	1002
LHS seed	–	42	42
Margin fraction	–	0.25	0.25

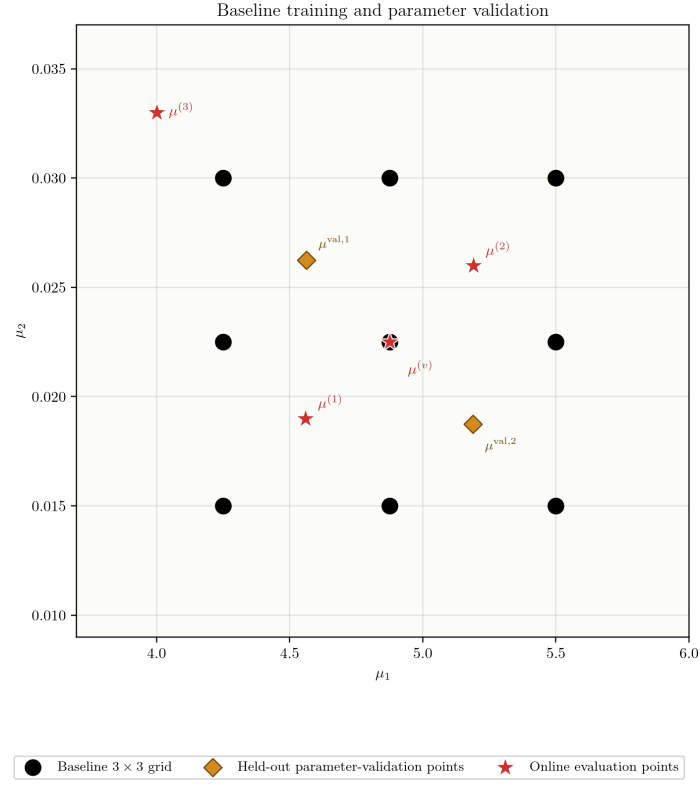


Figure 1: Baseline Euclidean-POD PROM data layout. Black circles are the nine HDM and linear-PROM training parameters; diamonds are the two held-out linear-PROM parameter-validation trajectories; red stars are the four online evaluation points.

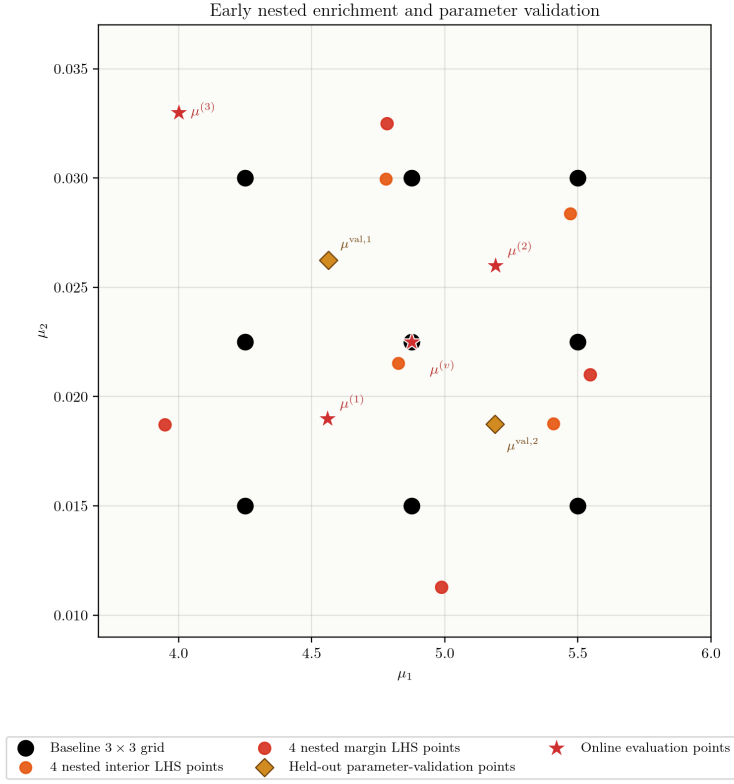


Figure 2: Early $9 + 8$ nested design. Four interior and four margin locations are added to the baseline grid. The extrapolatory point $\boldsymbol{\mu}^{(3)}$ lies in the upper-left corner, whereas the early subset samples its left and top margins separately.

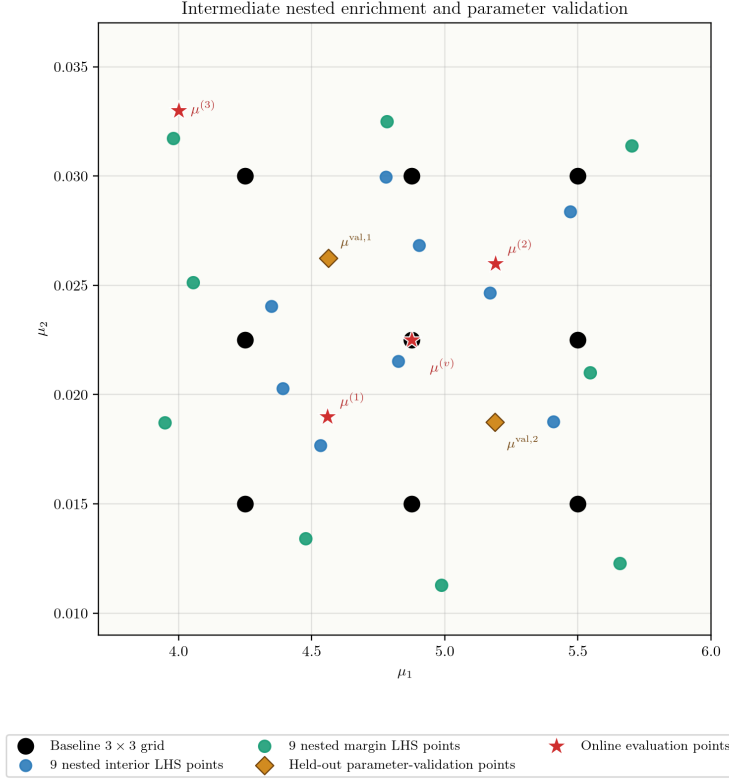


Figure 3: Intermediate 9 + 18 nested design. The added upper-left margin point near $(3.980, 0.03173)$ gives direct coverage close to $\boldsymbol{\mu}^{(3)} = (4.000, 0.0330)$.

4.4 Model training and shared Case-2 map

The baseline networks use 4509 training samples, and the 9 + 8 and 9 + 18 networks use 8517 and 13527, respectively. Every network uses the same 1002 external-validation parameter-time samples. Case 1 and Case 3 use $n = 10$ primary coordinates; PROM-POD-AE and POD-DL-ROM use $n_z = 10$. The architectures and reduced coordinate errors are reported in Table 3.

Table 3: Training and held-out parameter-validation errors in reduced coordinates. The master map is shared between the Case-2 tail and the POD-NN-ROM direct endpoint.

Model	Learned map	Train e_q (%)	Validation e_q (%)	Parameters
PROM-ANN Case 1	$\mathbf{q}_p \mapsto \mathbf{q}_s$	1.422	3.590	564,621
Master map (Case 2 / POD-NN-ROM)	$(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_{151}$	2.888	4.261	565,399
PROM-ANN Case 3	$(\mathbf{q}_p, \boldsymbol{\mu}, t) \mapsto \mathbf{q}_s$	0.572	3.482	565,389
PROM-POD-AE	$\mathbf{q}_{151} \mapsto \mathbf{z} \mapsto \hat{\mathbf{q}}_{151}$	0.194	2.673	486,817
POD-DL-ROM	$(\boldsymbol{\mu}, t) \mapsto \mathbf{z} \mapsto \hat{\mathbf{q}}_{151}$	1.843	2.981	952,747

For the controlled Case-2 comparison, one master map is trained as

$$\mathcal{G}_q : (\boldsymbol{\mu}, t) \mapsto \hat{\mathbf{q}}_{151}. \quad (34)$$

For primary dimension n , Case 2 inserts $\mathbf{q}_s = \mathbf{P}_{>n} \mathcal{G}_q(\boldsymbol{\mu}, t)$ into the residual solve, whereas POD-NN-ROM uses the entire vector directly. Thus $n = 0$ is exactly the POD-NN-ROM endpoint from this same trained map, and $n = 151$ is the linear PROM endpoint. The shared-map construction is a diagnostic control, not a restriction of the general Case-2 formulation.

4.5 Reported errors

For each evaluation parameter, the state error is

$$e_u(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F}{\|\mathbf{u}_{\text{HDM}}\|_F}, \quad (35)$$

and the coefficient error against the full linear PROM trajectory is

$$e_q(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{q}_{151}^{\text{lin}} - \hat{\mathbf{q}}_{151}\|_F}{\|\mathbf{q}_{151}^{\text{lin}}\|_F}. \quad (36)$$

The linear PROM is an accuracy reference for the fixed basis and full residual, not a universal lower bound with respect to the HDM.

5 Euclidean-POD PROM Results

5.1 Baseline state accuracy and solution fields

Table 4 reports space–time state errors. The linear PROM remains below 0.5% at the three in-domain points. Among the learned intrusive models, Case 3 and PROM–POD–AE remain closest to that reference in domain. Case 2 improves on the direct POD–NN–ROM endpoint at all four points, but its prescribed tail remains visible, particularly away from the training grid. POD–DL–ROM is the closer of the two non-intrusive baselines in this campaign.

Table 4: Online state errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. The mean averages the three in-domain points. The horizontal rule separates intrusive residual-minimizing models from non-intrusive maps.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear PROM	0.410	0.450	0.463	0.441	0.850
PROM–ANN Case 1	0.418	0.543	0.736	0.566	1.673
PROM–ANN Case 2 ($n = 10$)	1.163	1.900	1.758	1.607	2.696
PROM–ANN Case 3	0.408	0.512	0.563	0.494	1.521
PROM–POD–AE ($n_z = 10$)	0.411	0.533	0.711	0.551	1.754
POD–NN–ROM	1.187	1.918	1.989	1.698	4.351
POD–DL–ROM ($n_z = 10$)	0.827	1.178	1.370	1.125	2.267

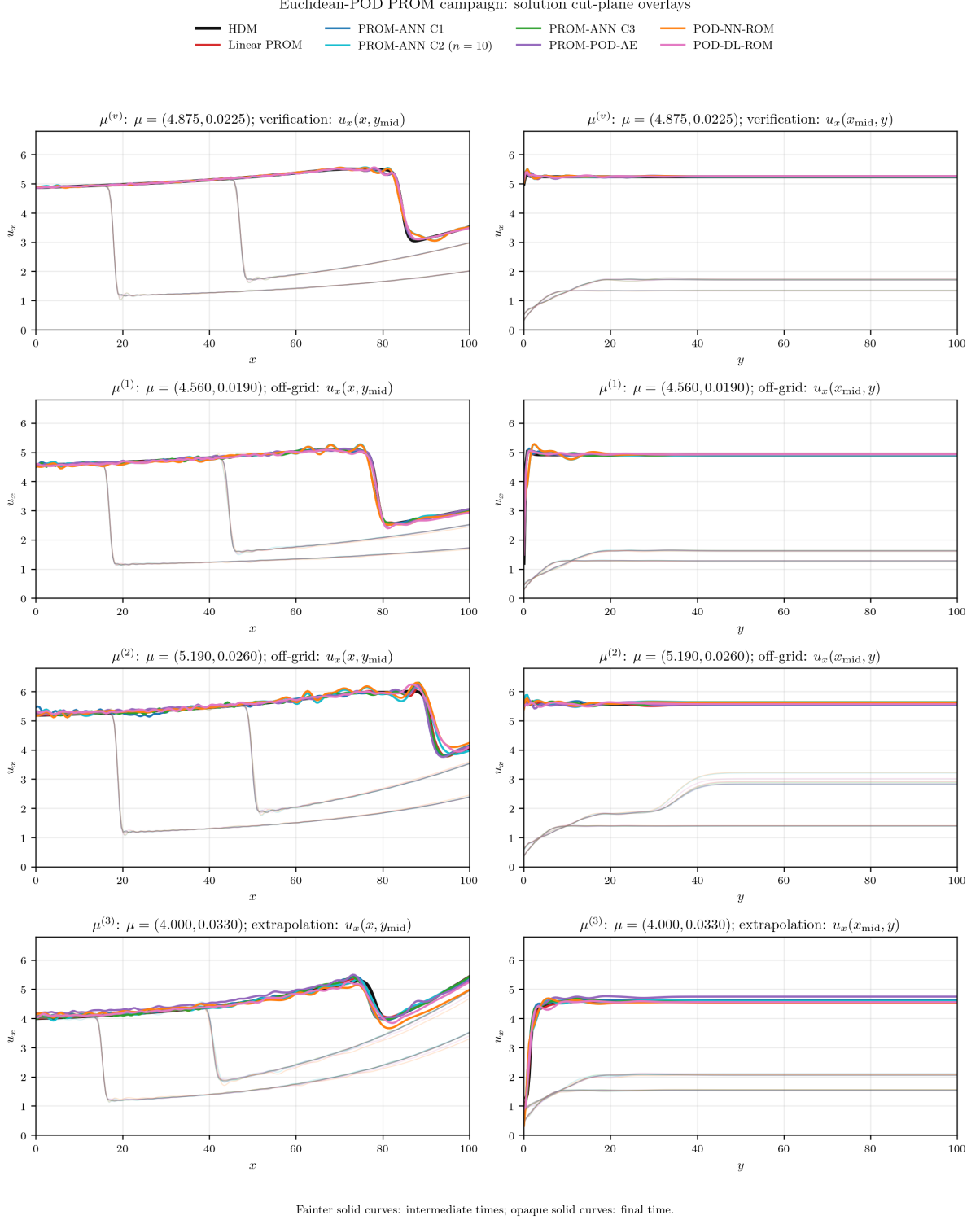


Figure 4: Final-time solution cut-plane overlays for the Euclidean-POD PROM campaign. Fainter solid curves show two intermediate times. Every intrusive curve uses the full residual; the two non-intrusive curves are direct reduced coordinate predictions.

5.2 Coefficient diagnostics

Table 5 and Fig. 5 compare each complete coordinate history with its 151-coordinate linear-PROM counterpart. The vertical line marks the $n = 10$ primary/secondary split used by

PROM-ANN. The direct maps are kept in the same figures and tables as the intrusive models, with an explicit separator, because all use the same linear-PROM coordinate teacher in this campaign.

Table 5: Relative coefficient-history errors e_q against the corresponding linear PROM. The horizontal rule separates intrusive and non-intrusive models, consistently with Table 4.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear PROM	0.000	0.000	0.000	0.000	0.000
PROM-ANN Case 1	0.324	0.751	1.391	0.822	3.304
PROM-ANN Case 2 ($n = 10$)	2.653	4.346	3.956	3.652	6.234
PROM-ANN Case 3	0.222	0.710	0.798	0.577	2.644
PROM-POD-AE ($n_z = 10$)	0.316	0.804	1.290	0.803	3.664
POD-NN-ROM	2.713	4.378	4.494	3.861	10.214
POD-DL-ROM ($n_z = 10$)	1.811	2.611	3.031	2.484	5.127

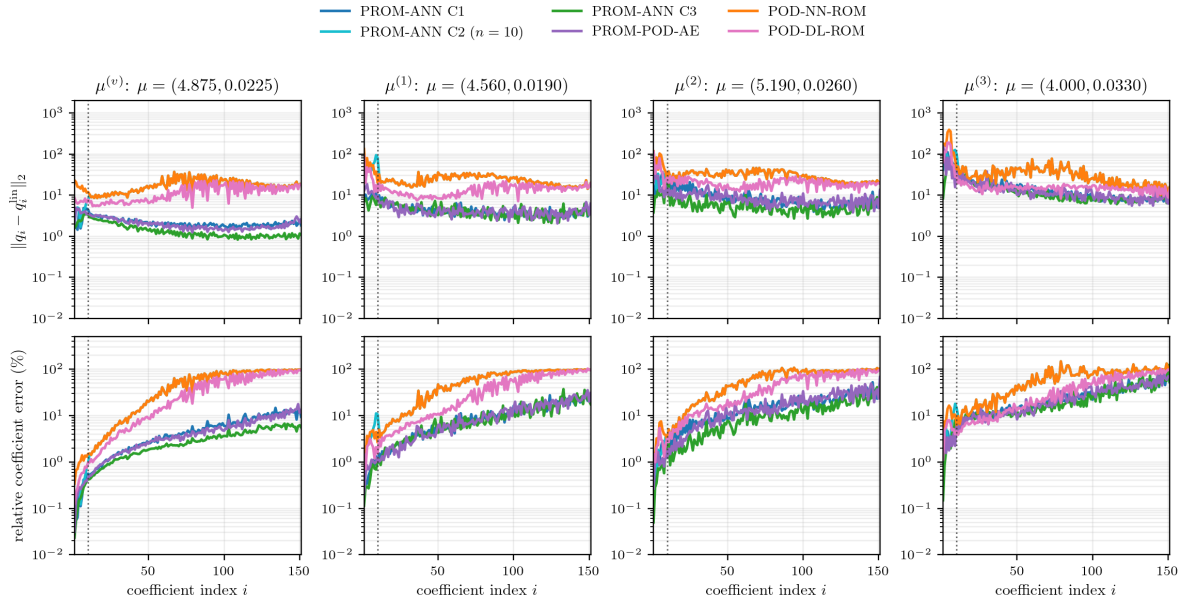


Figure 5: Per-coefficient absolute (top) and relative (bottom) errors against the 151-coordinate linear PROM. All panels in each row share the same logarithmic ordinate range. The dotted vertical line marks coefficient 10; all model curves are solid.

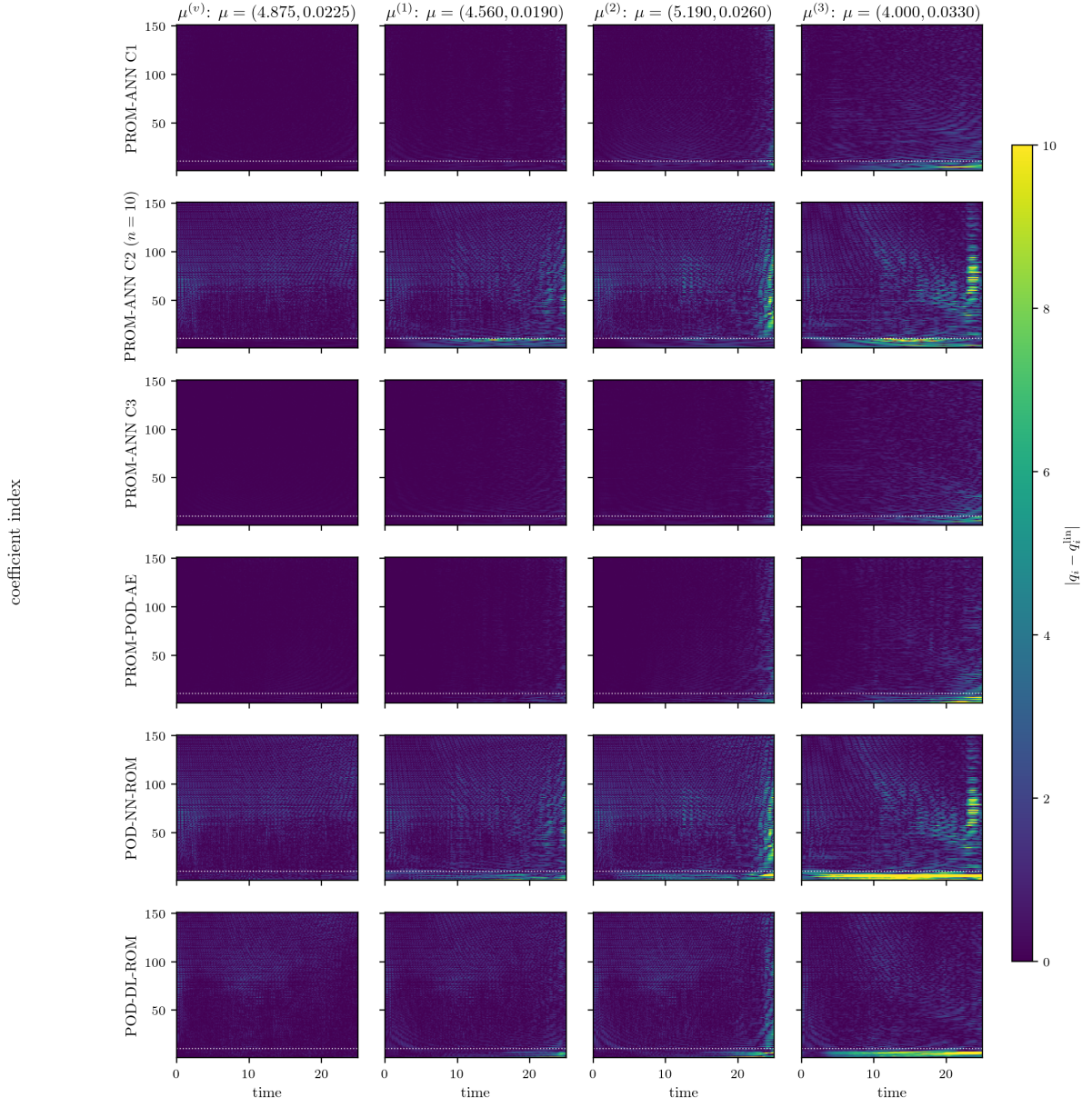


Figure 6: Absolute coefficient errors over time. Complete model names distinguish the intrusive PROM-POD-AE model from the non-intrusive POD-DL-ROM. The dotted white line marks the $n = 10$ primary split.

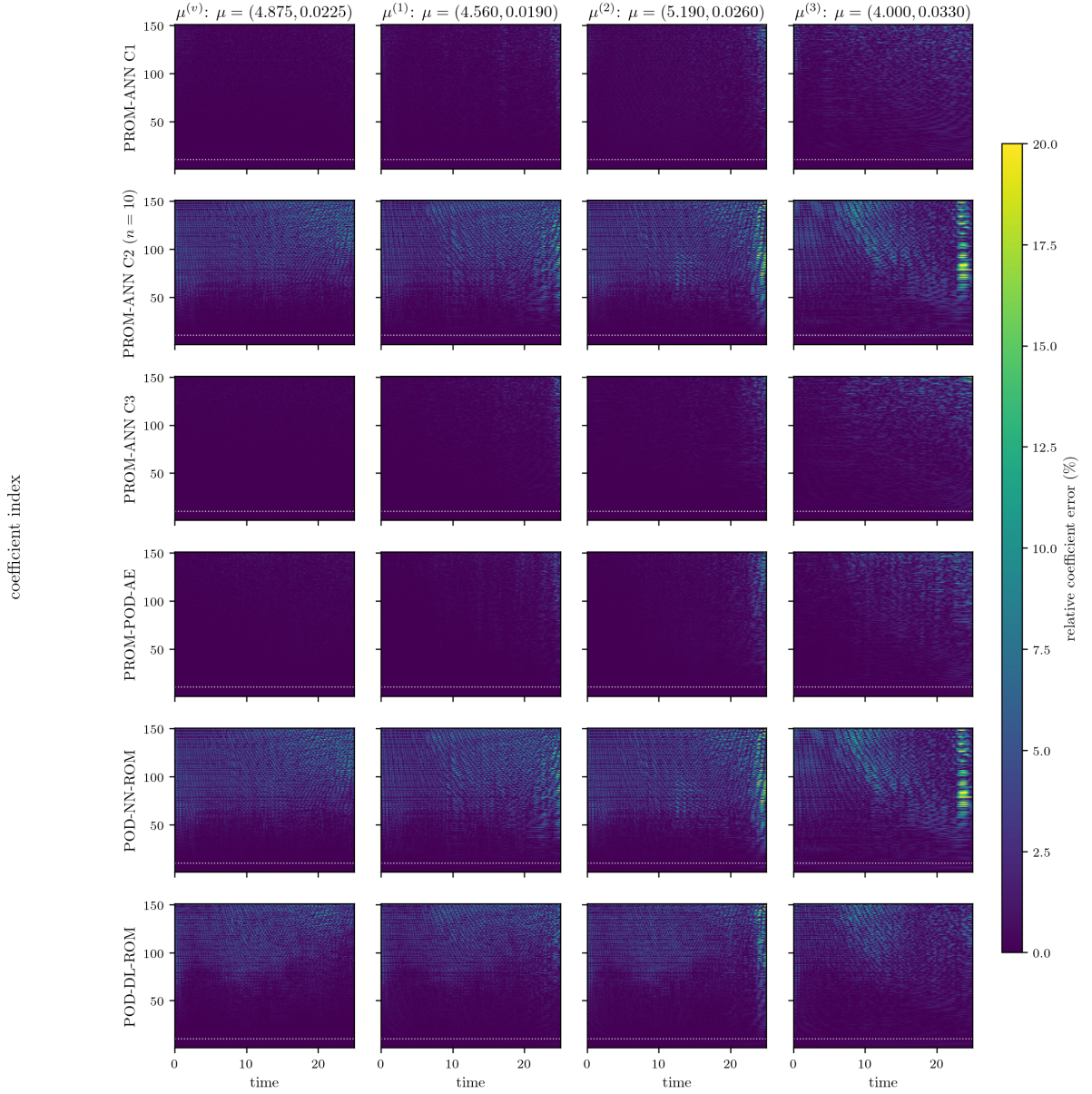


Figure 7: Relative coefficient errors over time, using the same ordering and coordinate split as Fig. 6.

5.3 Case-2 solved-dimension sweep

The shared master map permits an interpolation between a direct parameter-time regression and the full-coordinate linear PROM. Table 6 and Fig. 8 show that increasing n reduces state error, with the sharpest improvement between $n = 50$ and $n = 100$. This records the balance in this controlled Case-2 setting between coordinates supplied by the ANN and coordinates corrected by the residual.

Table 6: Case-2 state errors as the number of residual-solved primary coordinates is varied. The $n = 0$ and $n = 151$ endpoints are respectively the shared-map POD-NN-ROM and the linear PROM.

Solved n	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
0 (POD-NN-ROM)	1.187	1.918	1.989	1.698	4.351
10	1.163	1.900	1.758	1.607	2.696
20	1.158	1.556	1.665	1.460	2.201
30	1.154	1.529	1.601	1.428	2.178
50	1.161	1.506	1.486	1.384	2.036
100	0.785	0.958	0.863	0.869	1.183
151 (linear PROM)	0.410	0.450	0.463	0.441	0.850

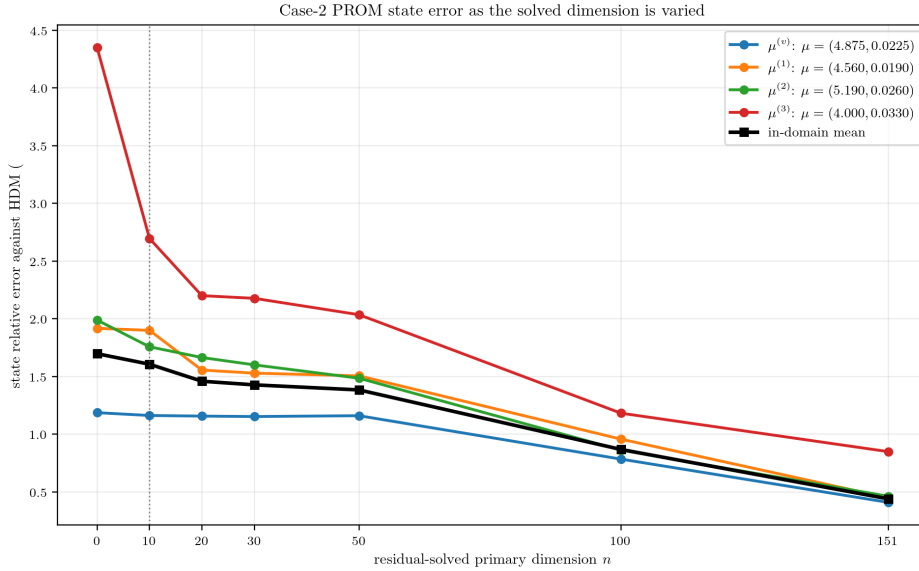


Figure 8: Case-2 state error versus the solved primary dimension. The black curve is the in-domain mean and the dotted line marks the baseline $n = 10$ split.

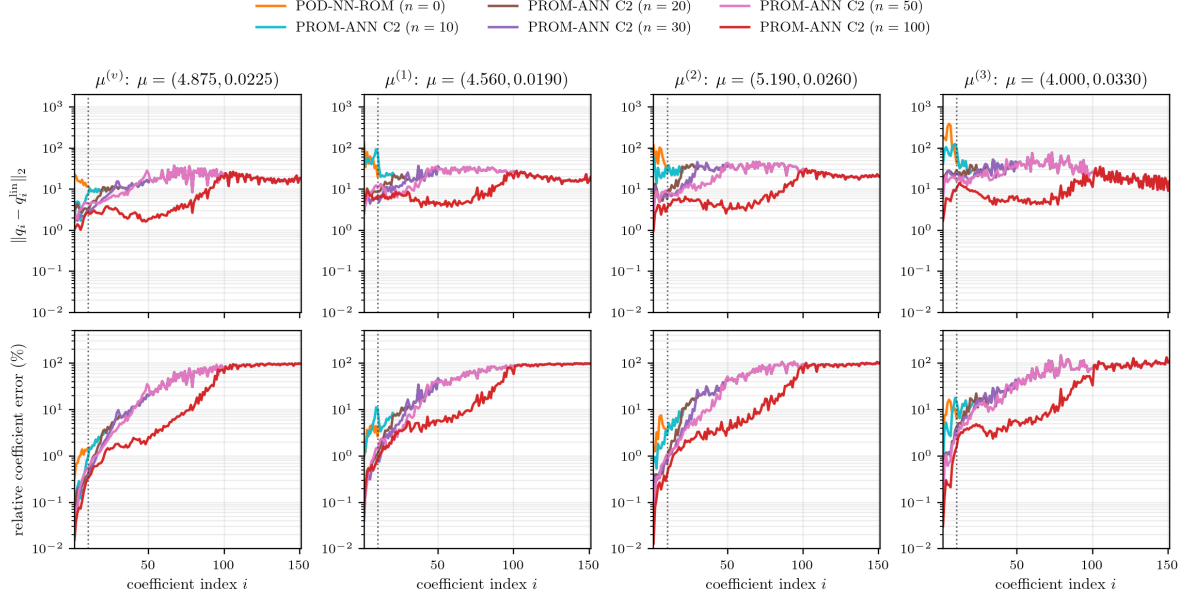


Figure 9: Case-2 coefficient trajectories in the shared-master-map sweep. Top: absolute error against linear PROM. Bottom: the corresponding relative error. The $n = 151$ linear-PROM reference has zero coefficient error and is therefore omitted.

5.4 Controlled Case-2 tail perturbation

To test the transfer mechanism independently of a regression magnitude, we start from the linear-PROM secondary history $\mathbf{q}_s^{\text{lin}}$ and add a perturbation in the direction of the actual master-map secondary error:

$$\mathbf{q}_{s\epsilon} = \mathbf{q}_s^{\text{lin}} + \frac{\epsilon}{100} \frac{\widehat{\mathbf{q}}_s - \mathbf{q}_s^{\text{lin}}}{\|\widehat{\mathbf{q}}_s - \mathbf{q}_s^{\text{lin}}\|_F} \|\mathbf{q}_s^{\text{lin}}\|_F. \quad (37)$$

For each prescribed percentage ϵ , Case 2 solves only the first ten coordinates. Thus the diagnostic uses the observed ANN-error direction but does not conflate its effect with a change of learned model. Table 7 and Fig. 10 show a monotone transfer from secondary-tail error to the recovered primary block. The amount of transfer is parameter dependent, so a tail loss alone does not imply a pointwise primary-coordinate bound.

Table 7: Primary-coordinate error induced by a controlled secondary-tail perturbation in Case 2 with $n = 10$. Each entry is relative to the corresponding linear-PROM primary history.

Imposed tail error (%)	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	$\mu^{(3)}$
0	0.001	0.001	0.001	0.000
1	0.131	0.095	0.060	0.098
3	0.382	0.283	0.169	0.291
5	0.617	0.469	0.265	0.480
10	1.133	0.922	0.453	0.940
15	1.544	1.364	0.602	1.389
20	1.856	1.801	0.773	1.839
30	2.281	2.702	1.413	2.778
50	4.034	4.949	4.208	5.044

Euclidean-POD Case-2 tail-perturbation diagnostic ($n = 10$)

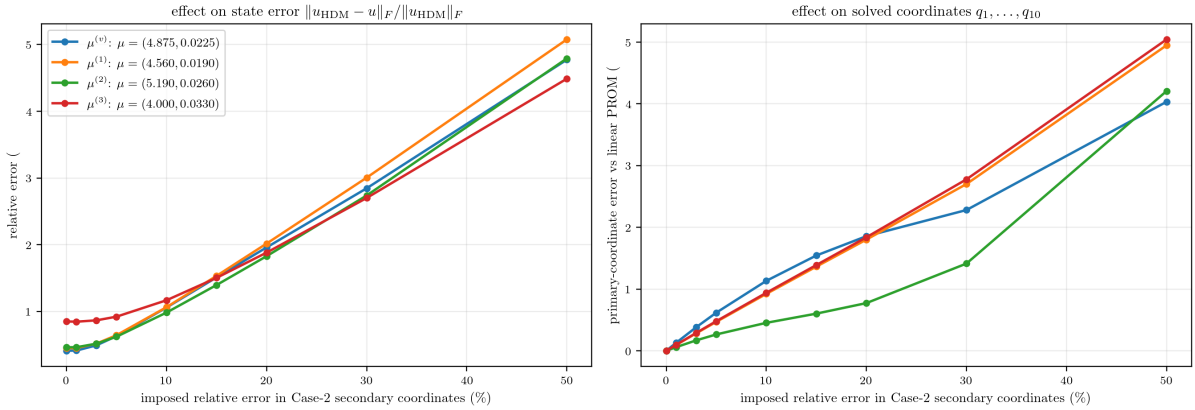


Figure 10: Euclidean-POD Case-2 sensitivity to a prescribed secondary-tail error, restricted to perturbations up to 50%. Left: state error against the HDM. Right: error in the ten residual-solved primary coordinates against linear PROM.

6 Effect of PROM Coefficient-Data Enrichment

The nested campaign tests whether the baseline discrepancies are primarily caused by the learned architecture or by parameter-space coverage. The Euclidean basis, network architectures, two parameter-validation trajectories, four online evaluation points, and full-residual PROM formulations are fixed. Only the number and location of linear-PROM coefficient-training trajectories change through the nested budgets 9, 9 + 8, and 9 + 18.

6.1 Offline regression accuracy

Table 8 reports errors in the reduced coordinates used to train and select the networks. All five learned families improve on the same external parameter-validation set as data are added.

The validation error of the master parameter–time map shared by Case 2 and POD–NN–ROM decreases from 4.261% to 1.607% with 9 + 8 and to 0.990% with 9 + 18. This monotone held-out improvement does not, however, imply monotone accuracy at a more distant extrapolation point. The three Euclidean master maps use the same wide SiLU architecture, raw MSE objective, external validation data, and random seed. Their selected checkpoints occur at epochs 1010, 1634, and 1731 for the baseline, 9 + 8, and 9 + 18 data, respectively, under the same validation-based early-stopping rule; none is the result of a wall-time interruption.

Table 8: Training and held-out parameter-validation coefficient errors across the Euclidean-POD PROM enrichment ladder. The master map supplies both the Case-2 secondary tail and the POD–NN–ROM direct prediction.

Model	Baseline (9)		Early (9+8)		Intermediate (9+18)	
	train e_q (%)	val. e_q (%)	train e_q (%)	val. e_q (%)	train e_q (%)	val. e_q (%)
PROM–ANN Case 1	1.422	3.590	0.830	1.208	0.504	0.634
Master POD–NN–ROM (Case 2 tail source)	2.888	4.261	0.882	1.607	0.577	0.990
PROM–ANN Case 3	0.572	3.482	0.255	0.610	0.284	0.452
PROM–POD–AE ($n_z = 10$)	0.194	2.673	0.164	1.192	0.144	0.488
POD–DL–ROM ($n_z = 10$)	1.843	2.981	0.224	0.433	0.211	0.249

6.2 Online state accuracy

Table 9 and Fig. 11 show a clear in-domain benefit. For Case 2 with $n = 10$, the mean state error falls from 1.607% to 0.702% with 9 + 8 and to 0.481% with 9 + 18. The direct POD–NN–ROM endpoint follows the same in-domain trend, from 1.698% to 0.735% and 0.484%. At 9 + 18, both are close to the fixed linear-PROM mean of 0.441%. Case 1, Case 3, PROM–POD–AE, and POD–DL–ROM improve more smoothly and are also close to the linear reference at the intermediate budget.

Table 9: Online state errors across the nested Euclidean-POD PROM coefficient-data ladder. Values are percentages. Each in-domain mean averages $\mu^{(v)}, \mu^{(1)}, \mu^{(2)}$; $\mu^{(3)}$ is extrapolatory. The linear PROM is unchanged, and the horizontal rule separates intrusive residual-minimizing models from non-intrusive direct maps.

Model	Baseline (9)		Early (9+8)		Intermediate (9+18)	
	in-domain mean	$\mu^{(3)}$	in-domain mean	$\mu^{(3)}$	in-domain mean	$\mu^{(3)}$
Linear PROM	0.441	0.850	0.441	0.850	0.441	0.850
PROM–ANN Case 1	0.566	1.673	0.460	1.129	0.439	0.850
PROM–ANN Case 2 ($n = 10$)	1.607	2.696	0.702	3.270	0.481	0.876
PROM–ANN Case 3	0.494	1.521	0.442	1.054	0.440	0.848
PROM–POD–AE ($n_z = 10$)	0.551	1.754	0.454	1.183	0.441	0.859
POD–NN–ROM	1.698	4.351	0.735	19.465	0.484	0.938
POD–DL–ROM ($n_z = 10$)	1.125	2.267	0.461	1.103	0.444	0.858

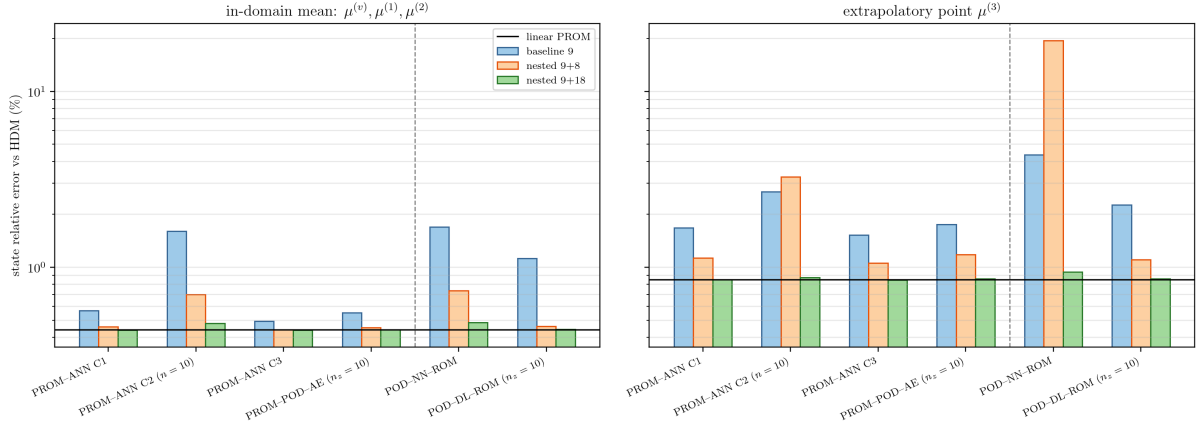


Figure 11: Effect of the three-level nested PROM coefficient-data ladder on online state error. Left: in-domain mean. Right: extrapolatory point. The two panels use exactly the same logarithmic ordinate limits, exposing the $9 + 8$ extrapolation anomaly without hiding the near-linear results. The dashed vertical rule separates intrusive PROMs from non-intrusive ROMs; the solid black horizontal level is the fixed linear PROM.

The extrapolatory result at $9 + 8$ is non-monotone and is retained rather than smoothed over. Case 2 changes from 2.696% at baseline to 3.270%, and POD-NN-ROM changes from 4.351% to 19.465%. Because these two models share the same parameter-time master map, while the other learned families improve at the same point, the degradation is specific to the extrapolation of that map rather than a uniform failure of the enriched data or the PROM solver. Residual minimization still provides substantial correction: using the same predicted 151-coordinate vector, Case 2 reduces the direct-map state error from 19.465% to 3.270% by solving the first ten coordinates.

The sampling geometry is consistent with this behaviour. The $9 + 8$ subset contains a left-margin point at approximately $(3.949, 0.0187)$ and a top-margin point at $(4.783, 0.0325)$, but no sample in their upper-left intersection near $\mu^{(3)} = (4.000, 0.0330)$. The $9 + 18$ subset adds $(3.980, 0.03173)$, after which the Case-2 and POD-NN-ROM extrapolatory state errors fall to 0.876% and 0.938%, respectively. This geometric observation explains why counting “margin” points alone is insufficient, but it does not establish that the particular $9 + 8$ deterioration is independent of optimization seed or stopping time.

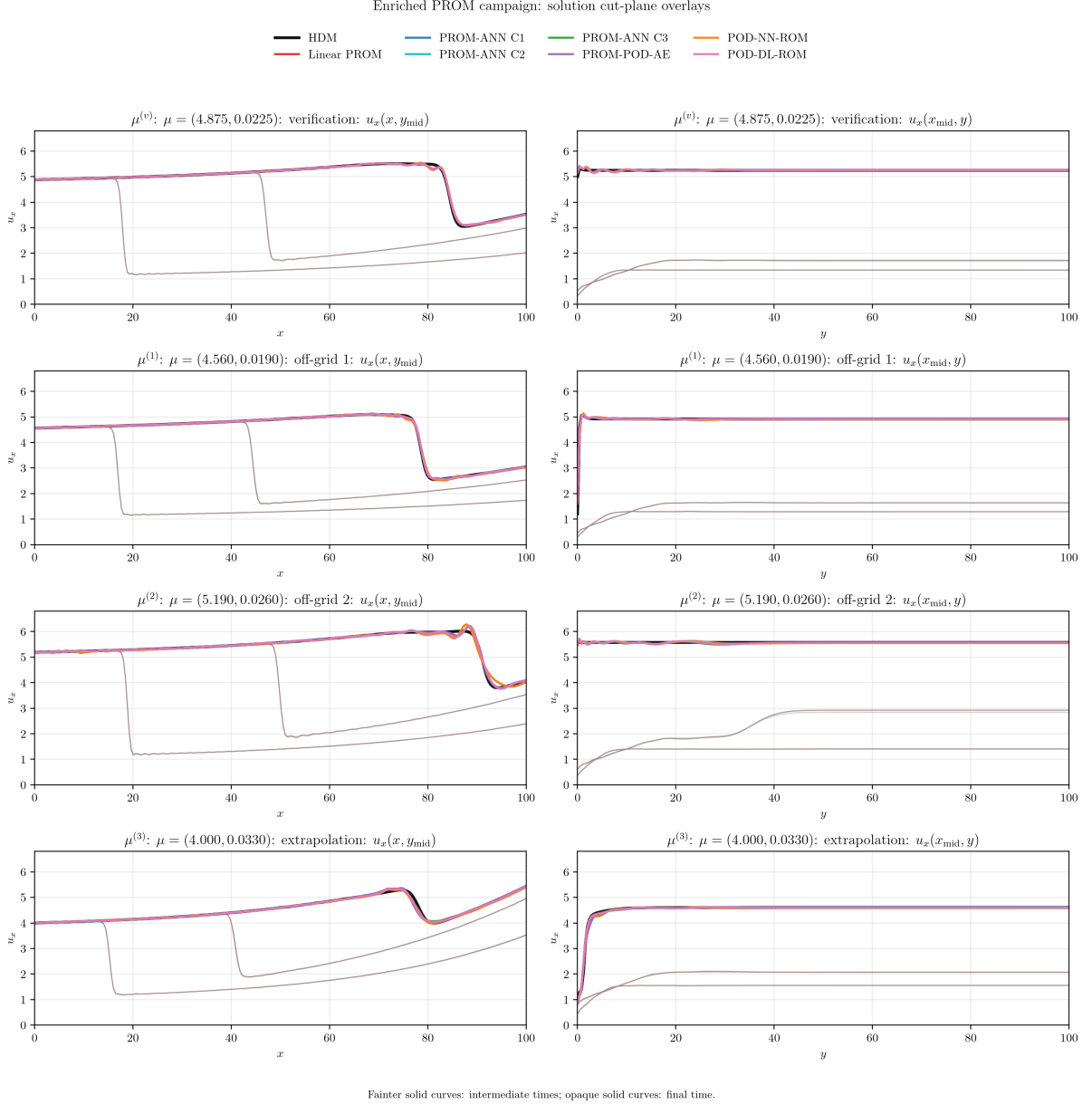


Figure 12: 9 + 18 Euclidean-POD PROM solution cut-plane comparison against the HDM. The model order, colours, state range, and time slices are identical to the baseline Fig. 4; all curves are solid.

6.3 Coefficient diagnostics

The coefficient results localize the non-monotone state behaviour before the online reconstruction. As shown in Table 10, the Case-2 extrapolatory coefficient error changes from 6.234% to 7.387% at 9 + 8, while the shared direct POD-NN-ROM error changes from 10.214% to 45.677%. The latter therefore already contains a severe coefficient-space extrapolation error. At 9 + 18, these errors fall to 0.589% and 1.005%. The state and coefficient evidence consistently identify the master-map extrapolation, rather than the Case-2 residual solve, as the source of the early-subset anomaly.

Table 10: Online coefficient-history errors against the fixed $n_{\text{tot}} = 151$ linear PROM across the nested data ladder. Values are relative Frobenius percentages over all coefficients and time steps. The horizontal rule separates intrusive and non-intrusive models.

Model	Baseline (9)		Early (9+8)		Intermediate (9+18)	
	in-domain mean	$\mu^{(3)}$	in-domain mean	$\mu^{(3)}$	in-domain mean	$\mu^{(3)}$
Linear PROM	0.000	0.000	0.000	0.000	0.000	0.000
PROM-ANN Case 1	0.822	3.304	0.317	1.571	0.122	0.208
PROM-ANN Case 2 ($n = 10$)	3.652	6.234	1.361	7.387	0.573	0.589
PROM-ANN Case 3	0.577	2.644	0.133	1.282	0.064	0.137
PROM-POD-AE ($n_z = 10$)	0.803	3.664	0.329	2.004	0.130	0.283
POD-NN-ROM	3.861	10.214	1.438	45.677	0.579	1.005
POD-DL-ROM ($n_z = 10$)	2.484	5.127	0.354	1.819	0.189	0.331

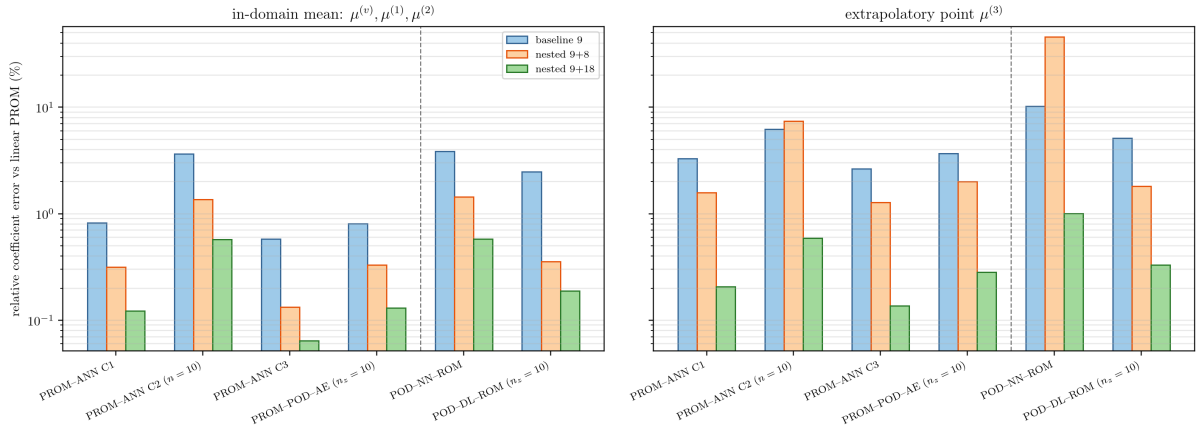


Figure 13: Effect of coefficient-data enrichment on global coefficient error. The logarithmic ordinate keeps both the $9 + 8$ POD-NN-ROM extrapolation error and the near-linear $9 + 18$ results visible. The dashed vertical rule separates intrusive PROMs from direct ROM maps.

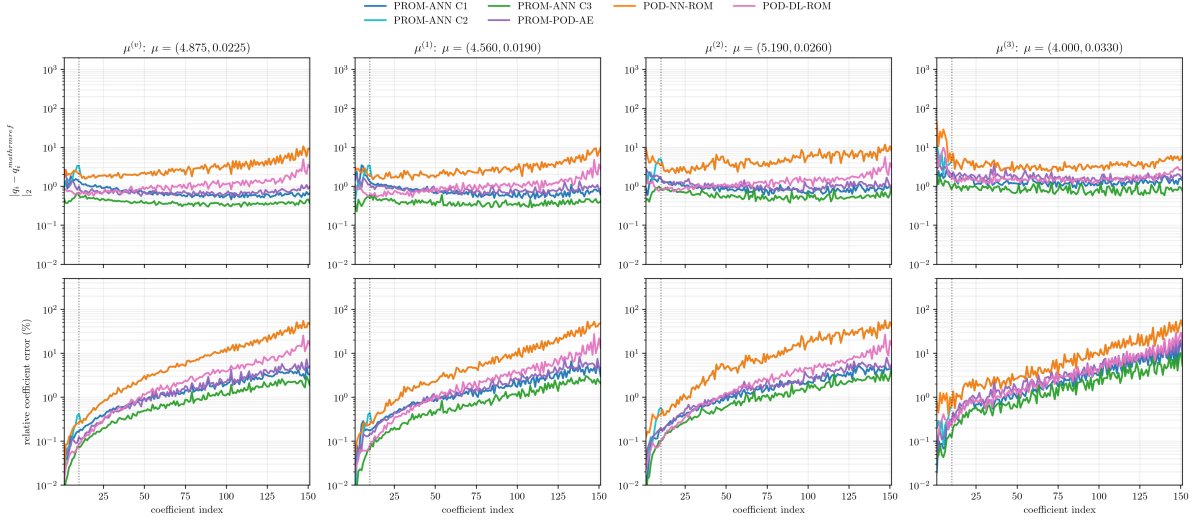


Figure 14: 9+18 per-coefficient absolute and relative trajectory errors against the linear PROM. The plotted set and fixed ordinate limits are identical to the baseline Fig. 5. The dotted vertical line marks the ten-coordinate primary/secondary split.

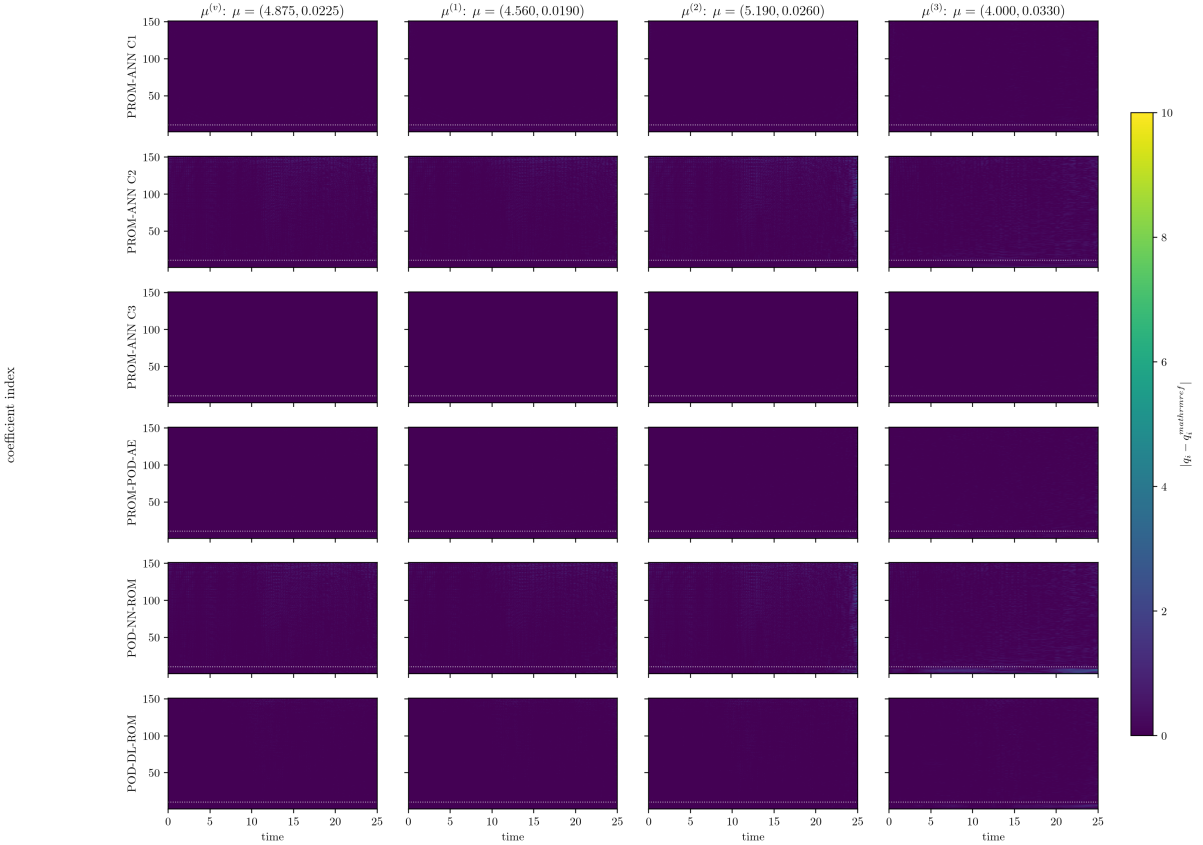


Figure 15: 9 + 18 time-by-coefficient absolute errors against the linear PROM. Rows contain the four intrusive models followed by POD-NN-ROM and POD-DL-ROM; the colour range matches the baseline diagnostic.

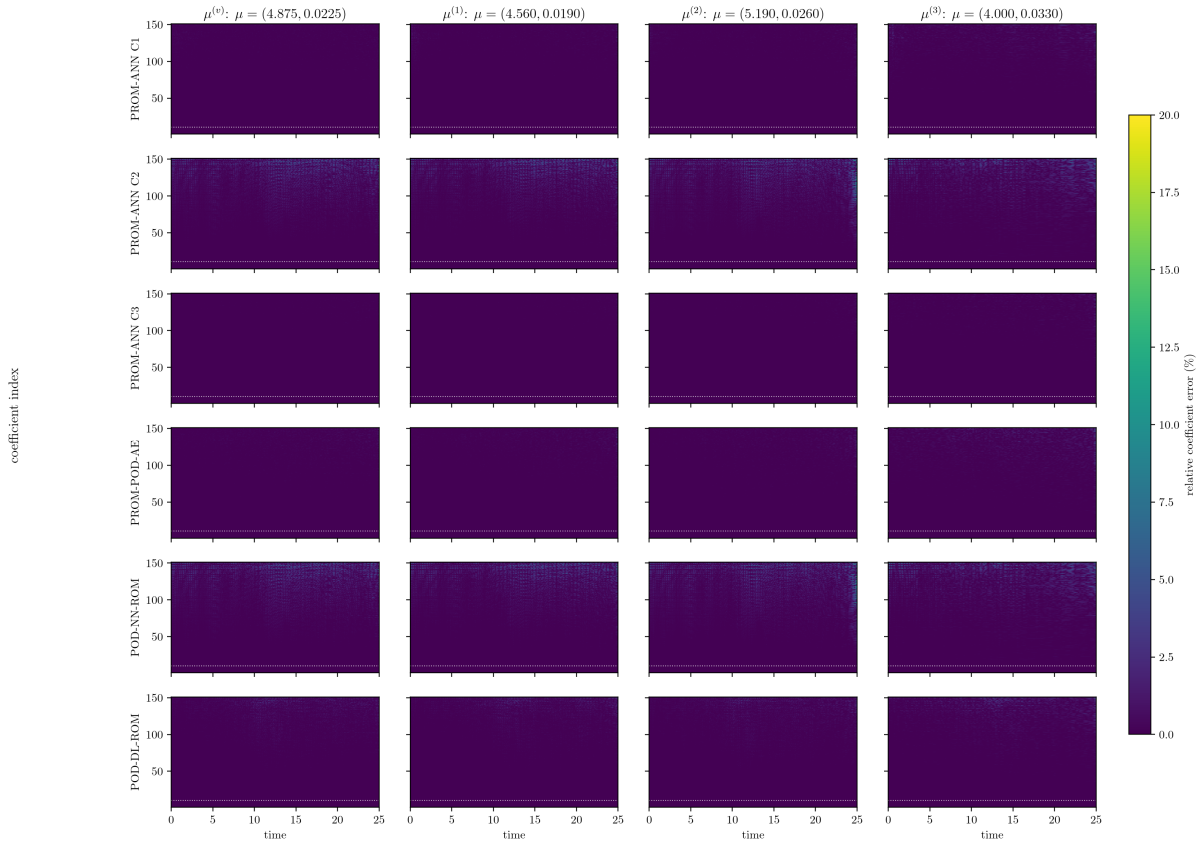


Figure 16: 9 + 18 time-by-coefficient relative errors, with the same common 0–20% colour range and model ordering as the baseline diagnostic.

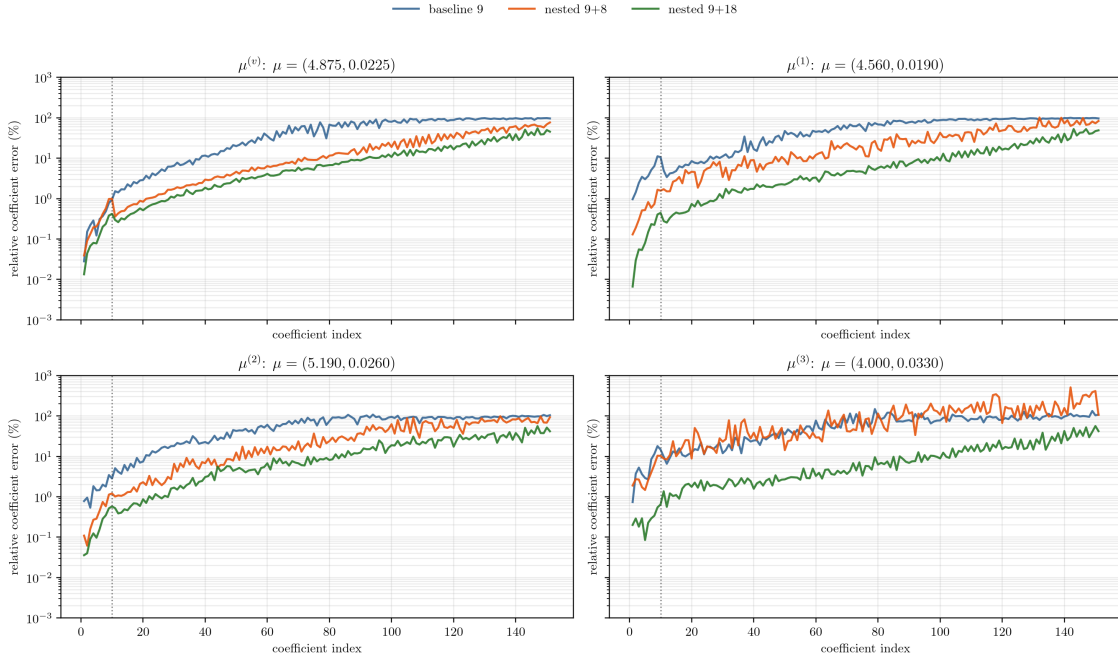


Figure 17: Case-2 per-coefficient relative error for the baseline, 9 + 8, and 9 + 18 master maps. Each panel is one online evaluation point. The dotted vertical line separates the ten residual-solved primary coordinates from the 141 ANN-injected secondary coordinates.

The nested experiment therefore supports two distinct conclusions. First, coefficient-data coverage is the dominant limitation in domain: already $9 + 8$ substantially improves every model, and $9 + 18$ brings all reported models close to the fixed linear-PROM reference. Second, enrichment size by itself does not control extrapolation. A small nested subset can reduce held-out in-domain validation error while worsening a corner extrapolation if its geometry does not cover the relevant joint parameter direction.

7 Conclusions

The Euclidean-POD PROM campaign gives a common full-residual comparison of three learned closures, an intrusive decoder, and two direct reduced-coordinate maps. The linear PROM remains the most accurate model in the fixed 151-coordinate trial representation. Case 3 and PROM-POD-AE retain close in-domain state accuracy while solving a low-dimensional online variable. The Case-2/POD-NN shared-map study gives a precise interpretation of intrusive correction: solving progressively more coordinates through the residual moves the result from the direct-map endpoint toward linear PROM, but the prescribed tail can still transfer its error into the solved block. The nested enrichment study shows that $9 + 18$ trajectories are sufficient to bring all learned models close to the linear-PROM reference at the reported points. The $9 + 8$ campaign improves every in-domain result but exposes a strong, localized master-map extrapolation failure, demonstrating that the geometry of added samples matters in addition to their number.

The results intentionally do not mix PROM evidence with an unavailable HPROM data set. The ECM derivation establishes the required model-specific construction: an HPROM rule must be built from each learned trial map and its tangent, rather than borrowed from a linear HPROM or from a different closure. The next HPROM campaign can therefore use the same reporting order, tables, and model comparisons as the PROM study.

A LSPG-Sensitive Basis Exploration

Before adopting the Euclidean POD basis used in the main text, we explored an M -regularized basis metric intended to emphasize sensitivity of the LSPG normal equations to reduced-coordinate perturbations. The objective was to reduce transfer from a learned secondary block to residual-solved primary coordinates without degrading the linear reference. In the present Burgers campaign, controlled tests did not establish a systematic improvement sufficient to justify making that metric the main experimental variable. For clarity and reproducibility, every reported result therefore uses the standard Euclidean POD basis of (4). The metric-aware construction is retained as an exploratory direction, not as a contribution or as a source of the numerical comparisons in this paper. The historical LSPG-sensitive enrichment campaign also selected an ELU master map, whereas the Euclidean campaign uses SiLU. Its earlier extrapolation trend is therefore not an architecture-controlled metric comparison and cannot be attributed to the basis choice alone.

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